

KYRA, Russian Federation

WMO: 309490

Lat: 49.567N

Lon: 111.967E

Elev: 908

StdP: 90.88

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.6	(c) -31.4	(d) -36.4	(e) 0.1	(f) -32.4	(g) -34.2	(h) 0.2	(i) -30.5	(j) 6.4	(k) -15.1	(l) 5.5	(m) -14.7	(n) 0.1	(o) 0	(p) 0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	29.7	16.4	27.5	15.9	25.6	15.2	18.7	25.3	17.8	23.9	16.9	22.6	1.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	13.4	20.9	15.8	12.5	20.0	14.8	11.7	19.2	57.4	25.1	54.1	24.0	51.2	22.6	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.7	(b) 5.5	(c) 4.5	(e) -35.8	(f) 33.1	(g) 2.7	(h) 2.8	(i) -37.7	(j) 35.1	(k) -39.3	(l) 36.8	(m) -40.9	(n) 38.4	(o) -42.8	(p) 40.4		
(a) 6.7	(b) 5.5	(c) 4.5	(e) -35.9	(f) 20.4	(g) 2.7	(h) 1.0	(i) -37.9	(j) 21.2	(k) -39.5	(l) 21.7	(m) -41.0	(n) 22.3	(o) -42.9	(p) 23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-20.7	-16.0	-6.9	2.5	9.7	16.0	18.5	16.3	9.7	0.4	-11.3	-18.9
	DBStd	14.55	4.76	5.39	5.74	4.50	4.23	3.42	2.61	3.11	3.96	4.80	6.28	5.62
	HDD10.0	4385	952	729	526	229	59	2	0	1	53	297	640	897
	HDD18.3	6752	1211	962	784	476	269	83	29	77	261	554	890	1156
	CDD10.0	737	0	0	0	3	48	183	263	196	43	1	0	0
	CDD18.3	64	0	0	0	0	1	14	34	14	1	0	0	0
	CDH23.3	1017	0	0	0	1	70	313	386	220	27	0	0	0
	CDH26.7	287	0	0	0	0	18	99	109	56	4	0	0	0
Wind	WSAvg	1.4	0.9	1.1	1.5	2.0	2.2	1.4	1.1	1.1	1.3	1.3	1.1	1.0
Precipitation	PrecAvg	377	2	2	3	12	25	58	123	98	41	10	3	2
	PrecMax	577	14	8	11	35	92	138	268	192	119	37	16	8
	PrecMin	205	0	0	0	0	3	8	22	22	3	0	0	0
	PrecStd	86	3	2	3	10	20	34	54	46	27	9	4	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.1	3.5	12.6	21.1	29.5	33.3	33.1	30.7	26.6	18.4	6.8	-2.2
		MCWB	-7.5	-1.6	3.8	8.2	13.5	16.8	18.2	17.0	13.4	8.0	1.4	-4.9
	2%	DB	-8.1	-0.9	8.9	17.6	25.2	29.3	29.4	28.1	22.8	14.6	3.1	-5.7
		MCWB	-10.0	-4.5	2.2	7.6	12.0	15.5	17.3	16.5	12.5	6.4	-0.6	-7.7
	5%	DB	-10.5	-3.5	5.6	14.4	22.4	26.6	27.2	25.6	20.6	11.6	0.6	-8.1
		MCWB	-12.0	-6.4	0.0	5.9	10.8	14.7	16.9	15.8	11.7	4.8	-2.5	-9.7
	10%	DB	-12.8	-6.4	2.8	11.7	19.7	24.5	25.2	23.5	18.0	8.7	-1.8	-10.3
		MCWB	-14.0	-8.6	-1.7	4.3	9.4	14.1	16.5	15.3	10.7	3.2	-4.2	-11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.3	-1.5	4.4	9.8	14.5	18.6	20.2	19.9	15.6	8.8	1.5	-5.1
		MCDB	-5.1	3.1	11.9	19.5	25.6	28.1	27.7	26.3	21.7	17.4	6.1	-2.5
	2%	WB	-9.9	-4.3	2.2	7.9	12.8	17.0	19.1	18.3	13.9	6.7	-0.5	-7.6
		MCDB	-8.1	-1.0	8.6	16.6	23.9	24.9	25.9	24.0	20.1	14.0	3.0	-5.8
	5%	WB	-12.0	-6.3	0.2	6.2	11.3	16.0	18.2	17.3	12.7	5.0	-2.4	-9.7
		MCDB	-10.5	-3.5	5.3	13.8	20.6	23.7	24.5	22.8	18.7	11.2	0.4	-8.2
	10%	WB	-13.9	-8.5	-1.5	4.6	10.1	15.1	17.4	16.3	11.3	3.5	-4.1	-11.6
		MCDB	-12.9	-6.3	2.6	11.1	18.5	22.3	23.1	21.4	17.3	8.3	-1.8	-10.4
Mean Daily Temperature Range	5% DB	MDBR	12.3	15.7	14.8	14.1	14.9	13.9	11.8	12.7	13.9	13.1	12.1	11.2
		MCDBR	13.9	19.1	18.9	19.9	21.2	19.5	16.5	17.3	19.1	18.6	14.0	12.4
		MCWBR	12.3	15.7	12.7	11.6	10.3	7.8	6.7	7.2	9.4	10.9	10.6	10.8
	5% WB	MCDBR	13.8	19.0	18.2	18.6	18.8	15.1	13.0	13.2	16.0	17.4	13.7	12.4
		MCWBR	12.2	15.7	12.4	11.4	9.8	6.9	6.1	6.3	8.1	10.4	10.5	10.8
Clear-Sky Solar Irradiance	taub	0.250	0.275	0.314	0.377	0.443	0.440	0.406	0.377	0.332	0.296	0.268	0.252	
	taud	2.316	2.288	2.273	2.179	2.020	2.144	2.307	2.366	2.438	2.459	2.380	2.282	
	Ebn at Noon	816	875	894	864	817	822	848	858	868	845	786	751	
	Edh at Noon	67	89	108	134	164	146	123	110	92	75	63	60	
All-Sky Solar Radiation	RadAvg	1.72	2.90	4.43	5.42	6.04	6.06	5.67	5.10	4.15	2.83	1.76	1.32	
	RadStd	0.06	0.09	0.17	0.34	0.36	0.38	0.40	0.37	0.28	0.13	0.08	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.91	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	-1.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page